❖ Networking Fundamentals and CIDR Basics

❖ What is CIDR? (Classless Inter-Domain Routing)

➢ **CIDR** stands for **Classless Inter-Domain Routing**. It's a method for allocating IP addresses and routing Internet Protocol packets that CIDR replaces the old **class-based IP addressing system** (Class A, B, C) by allowing **more flexible IP address allocation**.

❖ What is IP is an address?

➢ An **IP address** (Internet Protocol address) is a unique identifier assigned to a device connected to a network. It allows devices to **communicate with each other** over the Internet or local networks.

- VIPv4 Address: 32-bit address divided into four 8-bit octets (e.g., 192.168.1.1)
- Written in dotted decimal format (e.g., 192.168.0.1)
- Each octet ranges from 0 to 255

Types of IP Addresses

Type	Description
Public IP	Routable over the internet
Private IP	Used within local networks
Static IP	Permanently assigned to a device
Dynamic IP	Assigned by DHCP and can change
Loopback IP	127.0.0.1 – Refers to the local system
APIPA	169.254.0.0/16 – Automatic Private IP Addressing

❖ What is a Subnet Mask?

➢ A Subnet Mask determines which portion of the IP address refers to the network and which part refers to the host.

➢ It masks the IP address to identify the network boundary.

Example:

- IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0

➢ **This means:**

✓ First 24 bits (255.255.255) → Network portion

✓ Last 8 bits (0) → Host portion

❖ Classes of IP:

- Class A /8 | 0-126 (127.0.0.0 - Localhost / loopback address) | N.H.H.H | 0-255.0-255.0-255 | 256*256*256 | 1C

- Class B /16 | 128 - 191 | N.N.H.H | 0-255.0-255 | 256*256 | 65536
- Class C /24 | 192 - 223 | N.N.N.H | 0-255 | 256
- Class D | 224 - 239 | Scitific and experimental use
- Class E | 240 - 255 | Future Use

❖ Relationship Between IP, Subnet Mask, and Network

- **IP Address** :- Identifies a **device (host)** on a network. **Example**: 192.168.1.10
- **Subnet Mask** :- Separates the IP into **network** and **host** parts. **Example**: 255.255.255.0
- **Network** :-A group of devices sharing the same network portion of the IP address **Example**: 192.168.10.0
- **Broadcast Address** :- A **broadcast address** is a special IP address used to **send data to all devices** on a particular network **Example**: 192.168.10.255
- **Usable Host Range:-** The **total number of hosts** in a subnet refers to the **total number of IP addresses available** in that subnet — including both usable and reserved addresses. (network + broadcast excluded)

❖ CIDR Notation :-

CIDR stands for Classless Inter-Domain Routing. It is a method of allocating IP addresses and routing Internet Protocol packets. **CIDR Notation** is a compact way of writing an IP address **along with its subnet mask**, using a **slash (/) followed by a number**.

❖ Table: CIDR to Subnet Mask

CIDR	Subnet Mask	Host Bits	Total IPs	Usable Hosts
/8	255.0.0.0	24	16,777,216	16,777,214
/16	255.255.0.0	16	65,534	65,534
/24	255.255.255.0	8	256	254
/25	255.255.255.128	7	128	126
/26	255.255.255.192	6	64	62
/27	255.255.255.224	5	32	30
/28	255.255.255.240	4	16	14
/29	255.255.255.248	3	8	6

❖ What is the difference between TCP and UDP?

➢ **TCP (Transmission Control Protocol)**

- **Connection-oriented**: A connection is established before data is exchanged (like a phone call).
- **Reliable:** Ensures all data is received and in the correct order**.**
- **It is a sloweer**

✓ **Use cases:**

- Web browsing (HTTP/HTTPS)
- Email (SMTP, IMAP, POP)
- File transfer (FTP)

➢ **UDP (User Datagram Protocol)**

- **Connectionless:** Sends data without establishing a connection (like sending a letter).
- **Unreliable:** No guarantee of delivery, order, or error checking.
- It is Faster and ideal for time-sensitive transmissions**.**

✓ **Use cases:**

- Video streaming
- Online gaming
- Voice over IP (VoIP)
- DNS queries

❖ Explain the significance of the OSI model in networking.

➢ **The OSI model helps standardize network communication into 7 layers**:- Physical, Data Link, Network, Transport, Session, Presentation, and Application.

➢ Enables different vendors and systems to **interoperate**

❖ What is Subnetting?

➢ **Subnetting** is the process of dividing a large IP network into smaller, more manageable **sub-networks**, or **subnets**.

❖ Explain the difference between IPv4 and IPv6.

➢ **IPv4** is 32-bit, supports 4.3 billion addresses, and uses dot-decimal notation (e.g., 192.168.1.1).

➢ **IPv6** is 128-bit, supports a vast number of addresses, and uses hexadecimal notation (e.g., 2001:db8::ff00:42:8329).

❖ What is the difference between a public and a private IP address?

- **Private IP** is used within local networks (e.g., 192.168.x.x).
- **Public IP** is used to connect to the internet**.**
- **Private IPs** are not routable on the internet**.**

❖ How many usable IPs are there in a /30 subnet?

➢ A /30 subnet provides **4 IP addresses total**, but **only 2 are usable** for hosts.
1] 1 network, 2] usable hosts 1 3] Usable host 2 4] broadcast

❖ What Is a Broadcast Address?

➢ A **broadcast address** is a special IP address used to send data to **all devices** on a network segment **simultaneously**.

❖ What’s the difference between a switch and a router?

- A **switch** connects **multiple devices** in a local network.
- A **router** connects **different networks** and manages traffic between them.